CSE4007 Artificial Intelligence - Assignment #3

Due: Friday, May 29, 2026, 11:59 PM

Submission Guidelines

The submission should include a code (***both link to the Colab with full-access permission and a .py file***) and a report that has answers to the questions and results. Compress all files into a single `.zip` archive and submit it along with your report. Marks will be deducted if the submission does not include the requested files. Implement **all the TODO sections** indicated in the assignment code. Any form of plagiarism (e.g., using ChatGPT, copying works from others or the Internet, etc.) will result in an immediate zero for this assignment. ***Use this link to download the assignment code***.

Note: On Google Colab (free T4 GPU), once you complete the implementation of the provided code, training the model typically takes around 30–40 minutes, while inference usually completes within a few seconds. Actual times may vary depending on runtime conditions.

Problem #1: Learning the New Token

Diffusion personalization refers to the task of teaching a diffusion-based model to capture a novel subject from only a handful of examples and then express that subject in diverse images by leveraging the models pre-existing visual knowledge. Textual Inversion is a representative approach to this diffusion personalization task. By supplying just 3-5 reference images of a new concept, we introduce a **placeholder token** whose embedding is fine-tuned while the rest of the model (text encoder, U-Net, VAE) remains frozen. This targeted update gives precise control over generation via natural language prompts.

For detailed explanations, please refer to the accompanying Jupyter Notebook and the original paper:

An Image is Worth One Word:
Personalizing Text-to-Image Generation using Textual Inversion

In this problem, you will implement the essential steps of Textual Inversion, training a placeholder token to capture a specific visual concept by optimizing only its embedding within the pre-trained diffusion framework:

a) Set up the tokenizer and handle the initializer token.

b) Configure the model so that only the necessary components are updated during training.

c) Complete the interior of the `training_function`.

Problem #2: Use & Evaluate

In this problem, you will experiment with various settings to evaluate how well Textual Inversion performs under different conditions.

a) Write code that generates images at regular step intervals during the denoising process.

b) Observe and describe what each image reveals–what is emphasized at the initial steps, what appears at the final steps, and any patterns you observe across timesteps.

c) Design and conduct experiments under modified conditions to probe failure cases of Textual Inversion, exploring variations such as different reference images or prompts, altered learning rates or numbers of training steps, and comparing outputs before training (initializer tokens) versus after training (placeholder tokens).

d) Synthesize your qualitative observations and any quantitative measurements to discuss the advantages and limitations of optimizing only the token embeddings.

e) Justify your conclusions with evidence drawn from your experiments.

Problem #3: Additional Experiments: Success/Failure Analysis and Model Improvement Proposals

In this problem, you are encouraged to freely experiment with various settings in order to analyze the behavior of the model, identify limitations, and explore ways to improve performance. For each experiment, provide both qualitative (e.g., visual inspection) and quantitative (e.g., CLIP scores) analysis.

a) Conduct an in-depth analysis of when the method succeeds versus fails under the conditions above, and document your findings.

b) Propose concrete strategies for model improvement (e.g., data augmentation, altered training procedures, modified loss functions).

c) Based on your proposals, design and implement experimental code, and report your results.

d) After aggregating your results, articulate your personal observations and conclusions.